FOOT PRINTING(Information Gathering)–Detailed Notes

Foot printing:

1. Foot printing is the systematic collection of publicly available (and optionally directly queried) information about a target (domain, network, web app, person, organization) to build an asset/profile map.
2. It's the first phase of any security assessment or attack lifecycle — and the best place for defenders to find and reduce exposure.

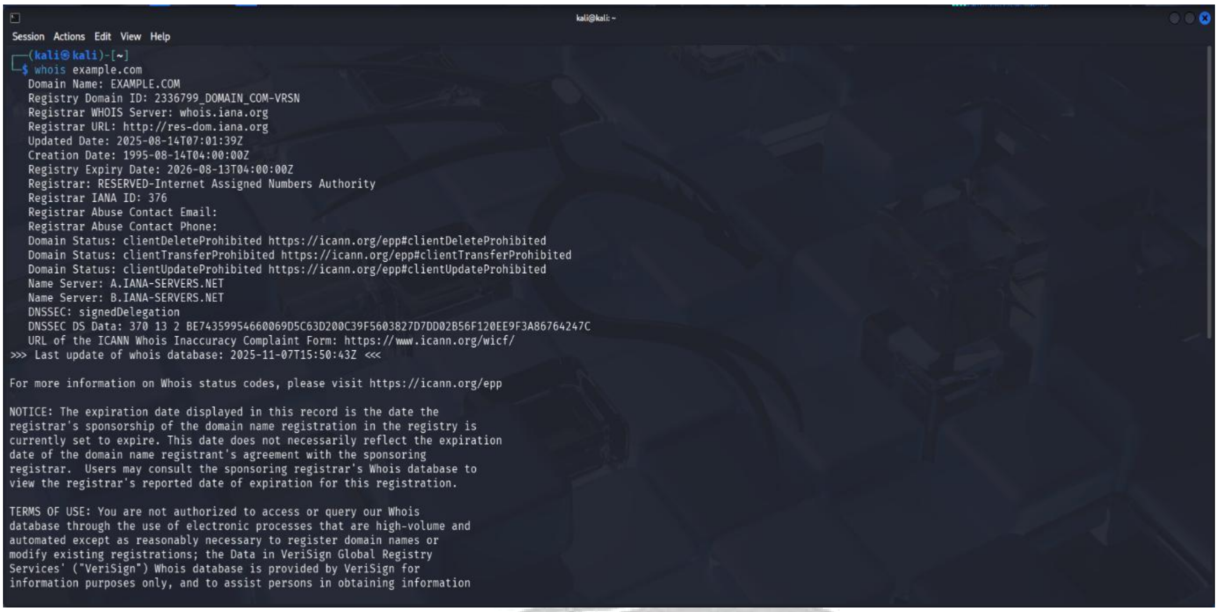
Type of foot printing:

1. Passive:
   - Collecting information about a target **without directly contacting the target's systems**.
   - You use publicly available sources (search engines, archives, public APIs, third-party indexes) so the target's infrastructure isn't queried and is less likely to detect you.
   - Build an initial picture of the target safely and legally; good for discovery and reconnaissance when you must avoid detection.
2. Active:
   - Gathering information by **directly interacting with the target's infrastructure or services**.
   - Examples: DNS queries to authoritative servers, HTTP requests, crawling, traceroute, or controlled port checks. These actions touch the target and are likely to be logged.
   - To confirm reachability, enumerate live hosts/services, discover hidden endpoints, or collect banners/version info that passive sources don't show.

Various step of information gathering:

1) WHOIS (domain registrar info)

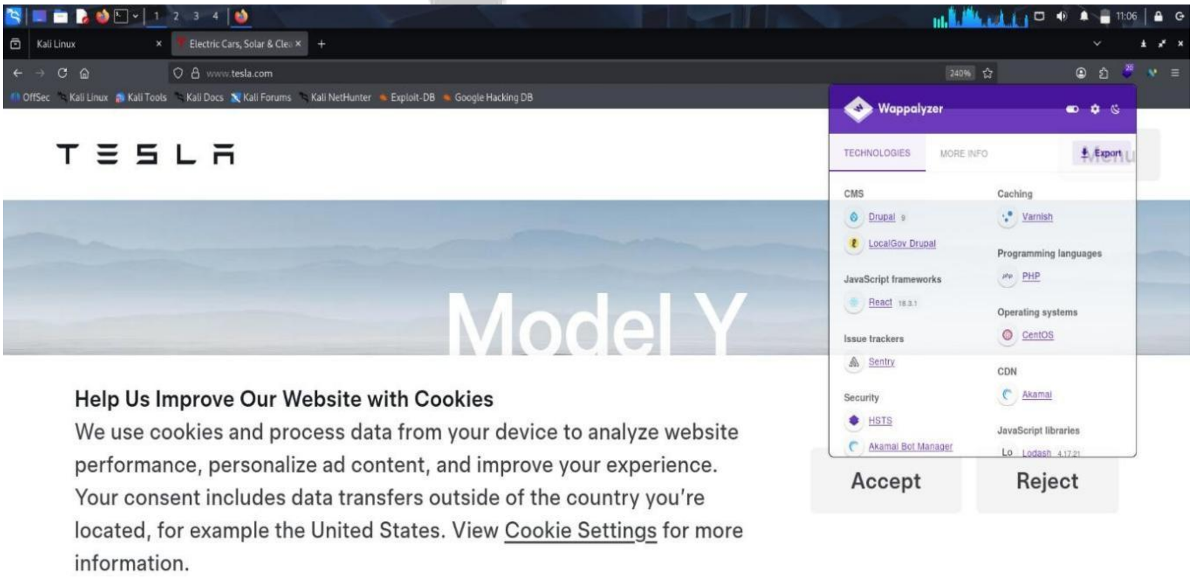
- **Goal:** Learn what WHOIS reveals and how to interpret registrar/expiry data.
- **Setup:** Create a fake test domain in a local hosts file (e.g., lab-example.local) and also use a public test domain you own OR use example.com for demonstration (passive read-only).


```
┌──(kali㉿kali)-[~]
└─$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
   Registrar URL: http://res-dom.iana.org
   Updated Date: 2025-08-14T07:01:39Z
   Creation Date: 1995-08-14T04:00:00Z
   Registry Expiry Date: 2026-08-13T04:00:00Z
   Registrar: RESERVED-Internet Assigned Numbers Authority
   Registrar IANA ID: 376
   Registrar Abuse Contact Email:
   Registrar Abuse Contact Phone:
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: A.IANA-SERVERS.NET
   Name Server: B.IANA-SERVERS.NET
   DNSSEC: signedDelegation
   DNSSEC DS Data: 370 13 2 BE74359954660069D5C63D200C39F5603827D7DD02B56F120EE9F3A86764247C
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-11-07T15:50:43Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
```

- **Observe:** registrar, creation/expiry, name servers, registrant contact (may be redacted).
- **Deliverable:** short note: registrar | created | expires | nameservers | privacy on/off.
- **Precautions:** WHOIS is passive; avoid querying personal/private domains without consent.

2) Wappalyzer / BuiltWith (tech fingerprinting)

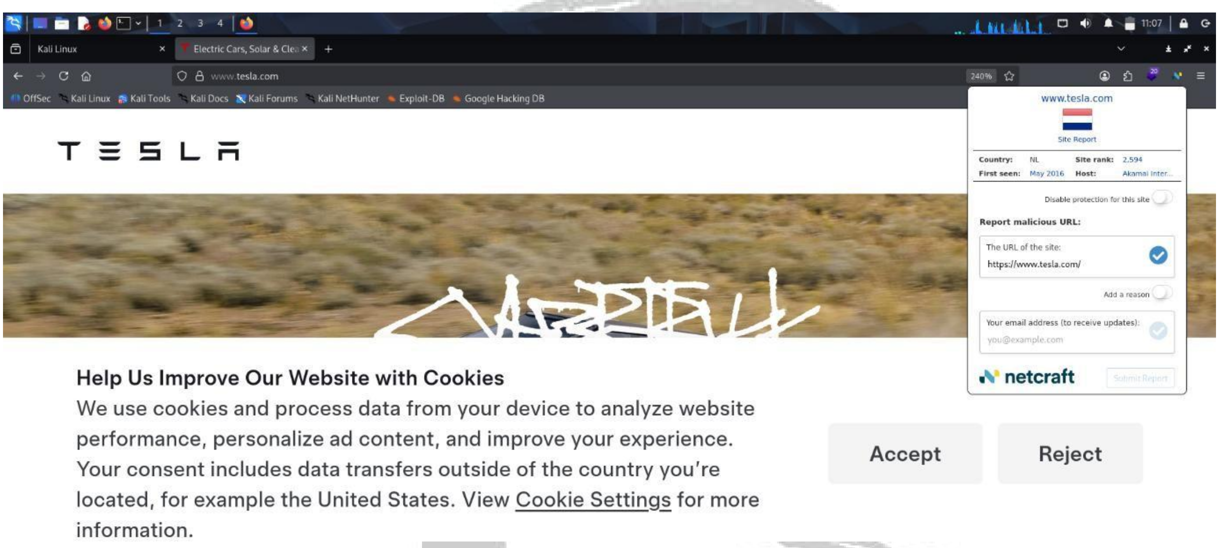
- **Goal:** Identify web technologies used by a target site.
- **Setup:** Run OWASP Juice Shop on target VM and open its web UI.
- Install Wappalyzer browser extension on your Kali desktop browser OR run:



- **Observe:** frameworks, server, JS libraries, analytics, CDN.
- **Deliverable:** table: component | evidence (header, script, cookie).
- **Precautions:** Passive detection — run only against lab targets.

3) Netcraft (historical/hosting info)

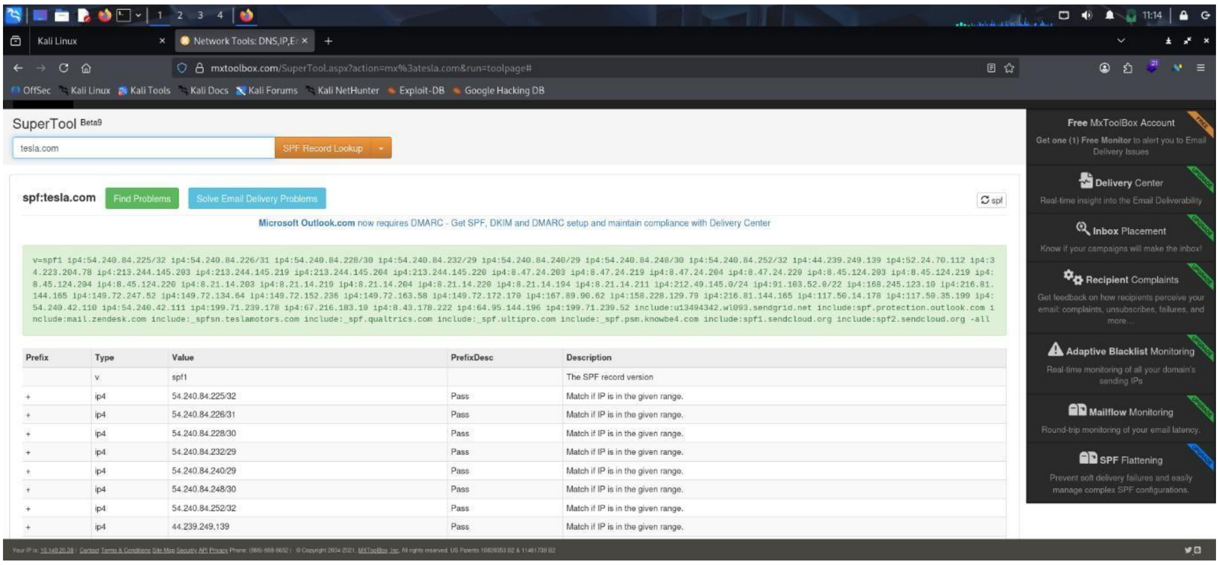
- **Goal:** See hosting/provider metadata and historical hosting if available.
- **Setup:** Use a public demo site you own OR use results from Netcraft for a public benign domain (only lookup).
- **Steps:** Visit Netcraft's site and enter the domain; note historical hosting, server header info.



- **Observe:** hosting provider, server/version, historical hosting changes.
- **Deliverable:** note of hosting provider + any interesting historical change.
- **Precautions:** Passive lookup only.

4) MX/DNS lookups (dig, nslookup, mxlookup)

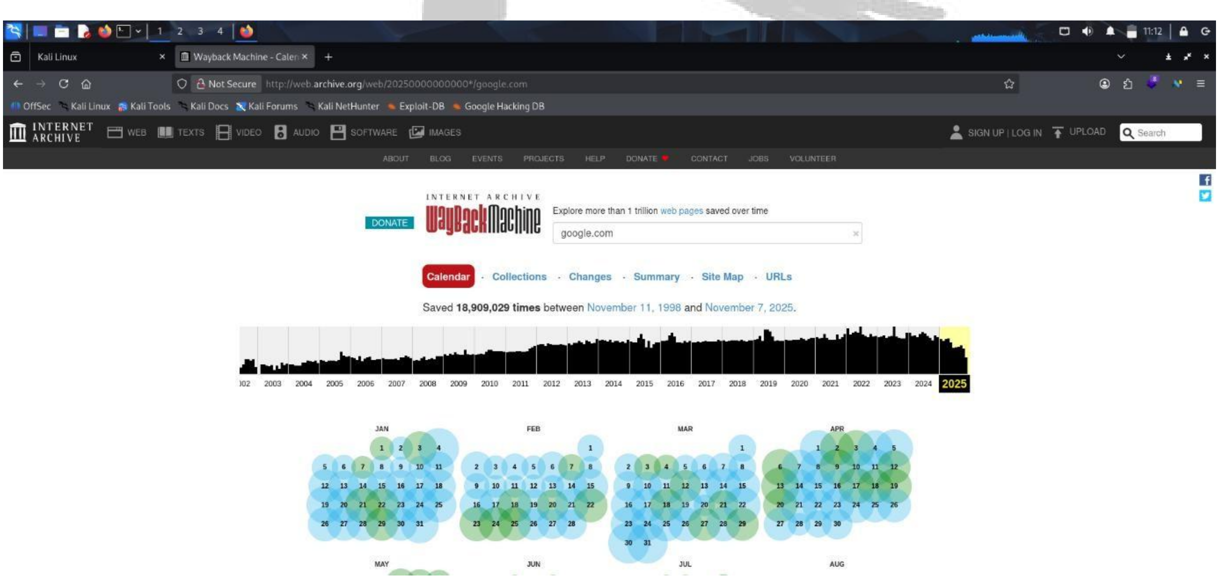
- **Goal:** Extract DNS records and mail configuration (MX, TXT/SPF, NS).
- **Setup:** Use a domain you control for full visibility. For lab practice use dig against public resolver for example.com.



- **Observe:** MX hostnames, SPF/DKIM/DMARC TXT entries, authoritative NS.
- **Deliverable:** DNS table: record type | value | notes (third-party services, misconfig).
- **Precautions:** DNS queries are low-impact but respect scope.

5) Wayback Machine (archive discovery)

- **Goal:** Discover historical pages/removed content that may expose endpoints.
- **Setup:** Use the Wayback Machine web UI for a benign public domain (or your own).
- **Steps:** Search https://web.archive.org/web/*/https://<your-domain> and browse snapshots.

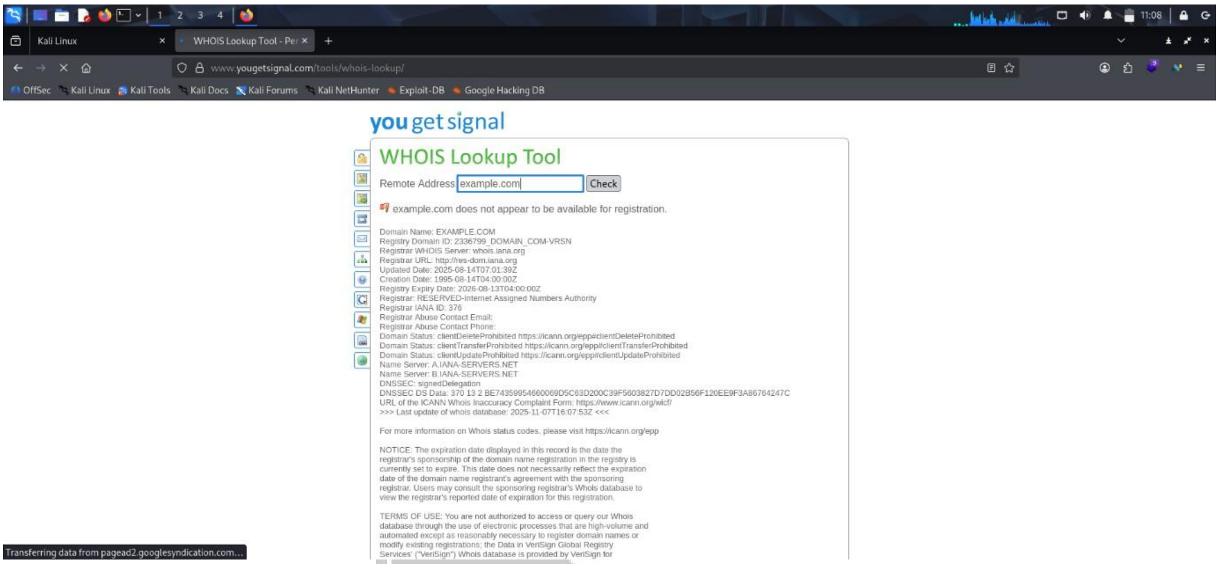


- **Observe:** old paths, previous pages, removed content, backup filenames.
- **Deliverable:** list of any sensitive-looking endpoints with snapshot date.

- **Precautions:** Passive only.

6) YouGetSignal-style reachability (simulate with nc / local checks)

- **Goal:** Understand remote port reachability; **do not** probe external hosts without permission.
- **Setup:** On target VM open a service (e.g., python -m http.server 8000) and from Kali test reachability.



- **Observe:** connection success/failure, open port presence.
- **Deliverable:** table: target IP | port | reachable | service.
- **Precautions:**Use local lab only. If using third-party web tools (YouGetSignal), be aware they perform remote checks.

7) GHDB (Google Dorks) & GitHub search

- **Goal:** Learn how Google dorks and GitHub searches reveal exposed files and credentials.
- **Setup:** Use Google with safe dorks on your own test repo or on public benign sites you manage.
- **Example dorks:**
- site:github.com "your-organization" "API_KEY"
- site:example.com filetype:pdf "confidential"
- **Observe:** exposed files, leaked config snippets, references to hosts/APIs.
- **Deliverable:** list of dorks used + results + remediation suggestions.

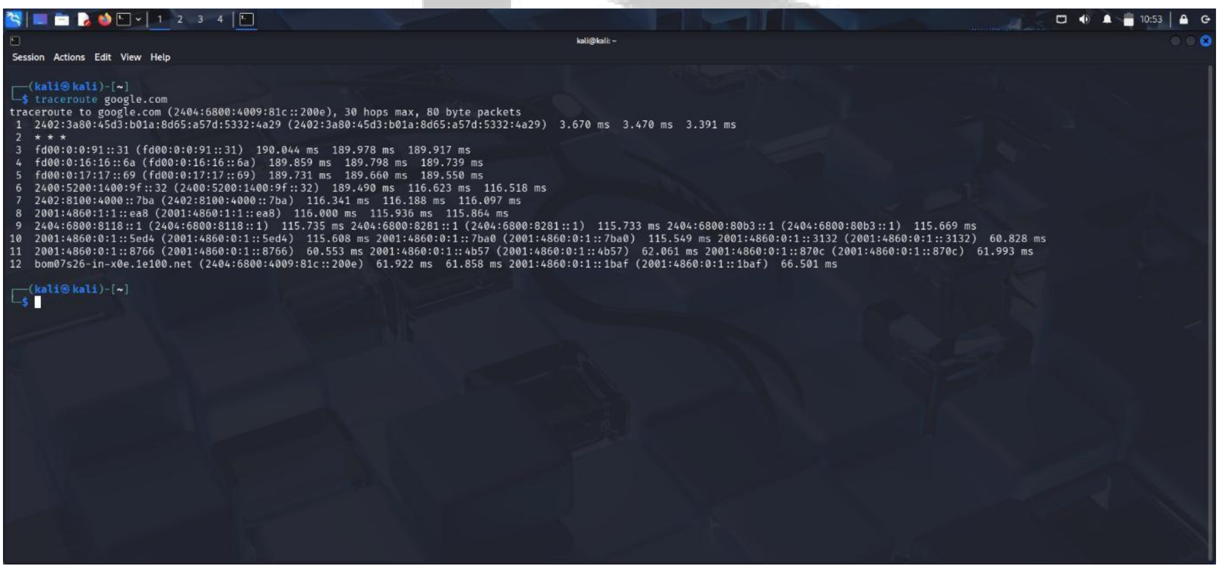
- **Precautions:** Do not use aggressive dorks on domains you don't control. Respect robots and privacy.

8) Shodan (internet-exposed service discovery)

- **Goal:** See what Shodan can reveal about exposed services (banners, ports).
- **Setup:** Use Shodan web UI or CLI against your lab public IP only if you expose it to internet (not recommended). Safer: use Shodan to search for your own public IPs that you control.
- **Example CLI:**
- shodan host <your-public-ip>
- **Observe:** service banners, ports, device type (if indexed).
- **Deliverable:** screenshot/export of Shodan host output + mitigation plan.
- **Precautions:** Don’t index or expose lab VMs to the public network unless intentional and secure.

9) traceroute / tracert (network path mapping)

- **Goal:** Map routing path from attacker to target and identify hops.
- **Setup:** Use local lab network.



- **Observe:** intermediate hops, latency spikes, possible NATs/routers.
- **Deliverable:** traceroute output + notes on network topology.

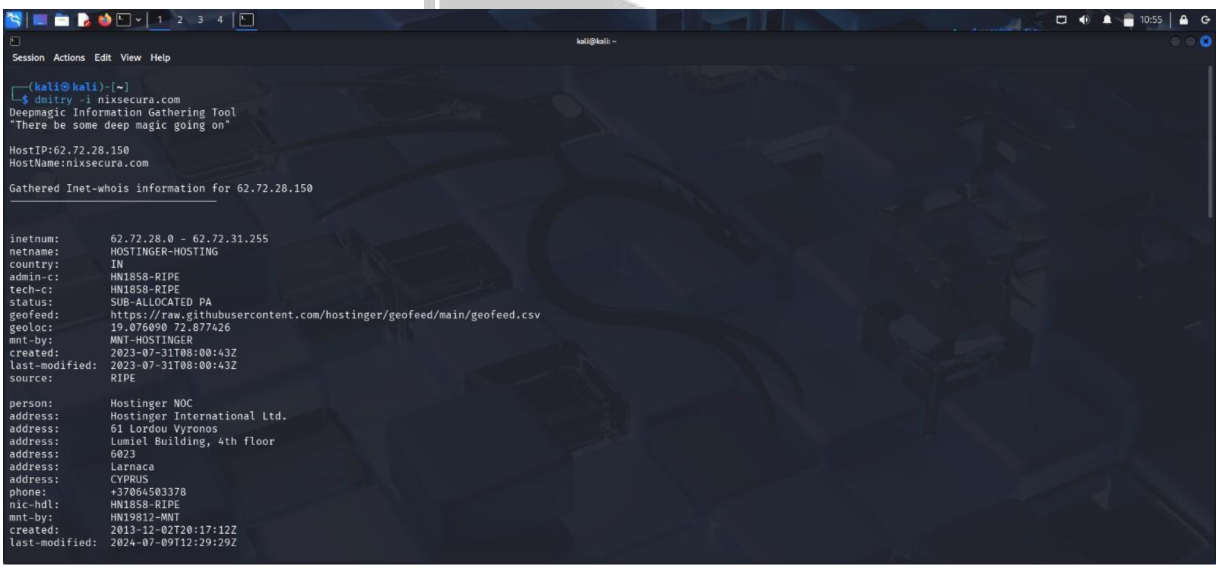
- **Precautions:** traceroute is low-risk; avoid using it for external reconnaissance without permission.

10) SpiderFoot (automated OSINT aggregator)

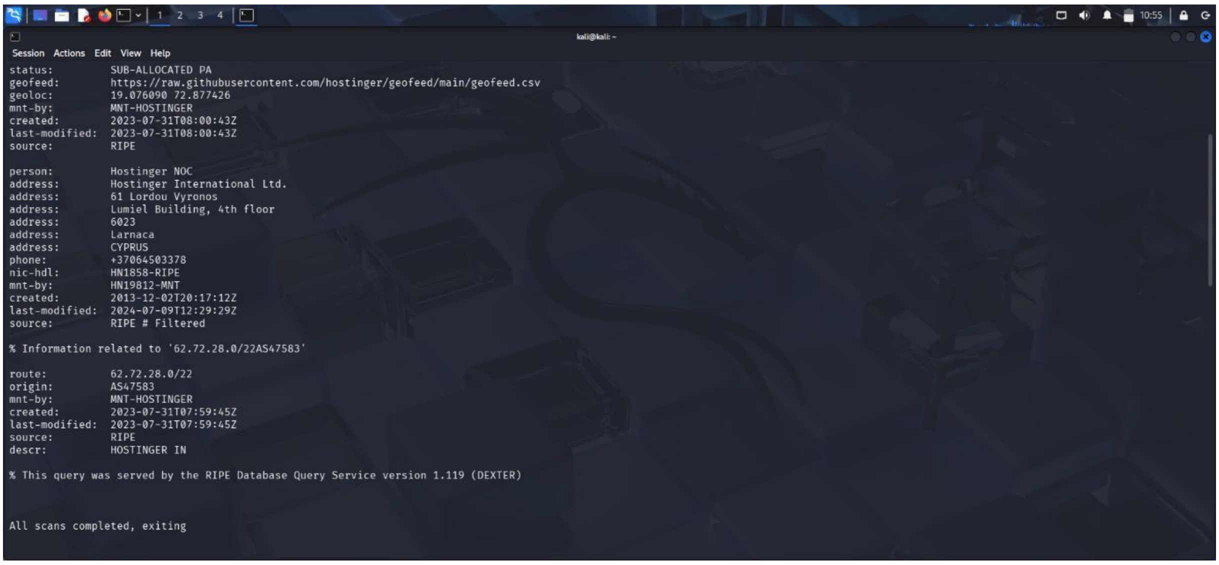
- **Goal:** Use SpiderFoot to aggregate passive OSINT into one report.
- **Setup:** Install SpiderFoot on attacker VM and configure with no active modules enabled unless authorized. Point at example.com or your own domain.
- **Example:**
- spiderfoot -s example.com -o spider_report.html
- **Observe:** subdomains, certs, public leaks, email addresses, third-party links.
- **Deliverable:** export HTML report + CSV of discovered assets.
- **Precautions:** Disable active modules (port scans, zone transfers) unless in-scope.

11) Dmitry & theHarvester (quick harvest tools)

- **Goal:** Compare outputs from small footprinting tools.
- **Setup:** Use lab domain.



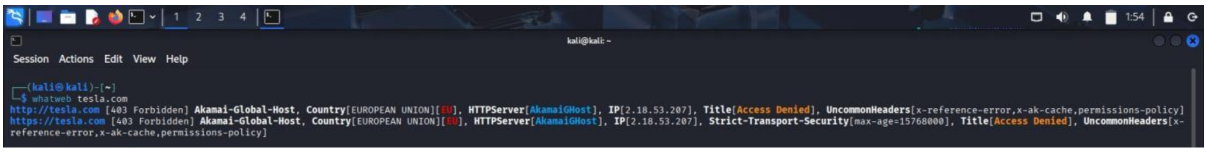
- **Observe:** email lists, subdomains, netcraft/whois results aggregated.
- **Deliverable:** combined list of emails/subdomains found and confidence level.
- **Precautions:** theHarvester uses public sources; be mindful of API limits and terms.



- **Goal:** Correlate all findings into a single asset inventory and prioritize.
- **Steps:** Merge outputs (WHOIS, DNS, subdomains, certs, services) into a spreadsheet with columns: asset | type | evidence | source | confidence | remediation.
- **Deliverable:** final spreadsheet + short slide with top 5 risky findings and mitigation.

12) Whatweb

- **Fast tech fingerprinting**: finds server types, CMS, frameworks, common plugins and sometimes version hints.
- **Prioritization**: helps you decide which follow-up checks to run (e.g., look for WordPress plugins, Node endpoints, or outdated server versions).
- **Defensive check**: defenders run it against their own sites to see what version/metadata they leak and what to harden.
- **Mostly active (low-noise)**: **WhatWeb** makes HTTP requests to the target and inspects responses.
- It's less noisy than a port scan but still *touches* the target, so it's active and will be logged by the target's web servers.
- Use it only on sites you own or have explicit permission to test.



13) wafw00f

- wafw00f is a small Python tool that detects and fingerprints Web Application Firewalls (WAFs) protecting a website.

- It starts with a normal HTTP request and, if needed, sends additional crafted requests to match signatures in its database.
- Use it to learn whether a site is behind a WAF and (often) which vendor/product is present.
- **Role in footprinting**: it’s an *active* web-fingerprinting step used after passive OSINT to confirm whether a target is protected by a WAF and to inform further (authorized) testing.